

# Exam 1

Name: Solutions

The exam contains 7 pages (including this cover page) and 6 questions. The total number of possible points is 50. You have 90 minutes to complete the exam.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **No outside resources allowed**. You may not use any notes, calculators, textbooks, or other resources on this exam. Make sure all electronic devices are silenced and put away.
- **Partial credit will be given**, so write down as much as you know even if some part of the proof might be missing.
- **Read the questions carefully**, make sure you answer the question being asked and understand the statement of the problem.
- **Answer all questions in the space provided**, you may use the back of pages as scrap paper, but only what is written under the question will be graded.

| Question | Points | Score |
|----------|--------|-------|
| 1        | 5      |       |
| 2        | 5      |       |
| 3        | 10     |       |
| 4        | 10     |       |
| 5        | 10     |       |
| 6        | 10     |       |
| Total:   | 50     |       |

Do not write in the table to the right.

1. (5 points) Consider the following subsets of  $\mathbb{R}$  and use them to answer the following questions. Write your answers in the boxes provided. **No justification is needed.**

•  $A = \{x \in \mathbb{R} : |x - 2| < 3\} = (-1, 5)$

•  $B = \{x \in \mathbb{R} : x^2 \leq 2\} = [-\sqrt{2}, \sqrt{2}]$

- (a) Express  $A \cap B$  as an interval in  $\mathbb{R}$ .

$$(-1, \sqrt{2}]$$

Find the following if they exist or indicate they do not exist:

- (b)  $\min(A \cap B)$

DNE

- (c)  $\max(A \cap B)$

$$\sqrt{2}$$

- (d)  $\inf(A \cap B)$

$$-1$$

- (e)  $\sup(A \cap B)$

$$\sqrt{2}$$

2. (5 points) Determine if each of the following statements is TRUE or FALSE by completely filling in the corresponding bubble. **No justification is needed.**

(a) The negation of the statement “ $x$  and  $y$  are both even integers” is “ $x$  and  $y$  are both odd integers”.

☐ TRUE

☒ FALSE

(b) All unbounded sequences are divergent.

☒ TRUE

☐ FALSE

(c) All divergent sequences are unbounded.

☐ TRUE

☒ FALSE

(d) If a set  $S$  has a maximum value, then  $\sup S \in S$ .

☒ TRUE

☐ FALSE

(e) The negation of the statement “for all real numbers  $a$ , there exists a real number  $b$  such that  $ab = 1$ ” is the statement “there exists a real number  $a$  such that for all real numbers  $b$ ,  $ab \neq 1$ ”.

☒ TRUE

☐ FALSE

3. (10 points) Use a **proof by contradiction** to prove the following statement. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $x \in \mathbb{R}$ , then  $x + \sqrt{2}$  is irrational or  $x - \sqrt{2}$  is irrational.

Assume, by way of contradiction, that  $x \in \mathbb{R}$  and  $x + \sqrt{2}$  is rational and  $x - \sqrt{2}$  is rational. Then, by definition of the rational numbers, there exist integers  $p, q, r, s$ , with  $q \neq 0$  and  $s \neq 0$  such that  $x + \sqrt{2} = \frac{p}{q}$  and  $x - \sqrt{2} = \frac{r}{s}$ . Then

$$2\sqrt{2} = (x + \sqrt{2}) - (x - \sqrt{2}) = \frac{p}{q} - \frac{r}{s} = \frac{sp - rq}{qs}$$

and  $\sqrt{2} = \frac{sp - rq}{2qs}$  where  $sp - rq$  and  $2qs \neq 0$  are integers by the closure of the integers under addition and multiplication. Thus  $\sqrt{2}$  is a rational number, which is a contradiction as we have proven in class that  $\sqrt{2}$  is irrational.

4. (10 points) Use a **proof by induction** to prove the following statement. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $n \in \mathbb{N}$  and  $n \geq 3$ , then  $2^n > n + 4$ .

Base case: When  $n=3$ ,  $2^3=8$ ,  $3+4=7$ .  $8 > 7$ .

Inductive step. Assume, for some  $k \in \mathbb{N}$ ,  $k \geq 3$ , that

$2^k > k + 4$ . Then

$$2^{k+1} = 2 \cdot 2^k > 2 \cdot (k+4) = 2k+8 > k+8 > k+5 = (k+1)+4$$

So if  $2^k > k+4$  then  $2^{k+1} > (k+1)+4$  and the

statement  $2^n > n+4$  holds for all  $n \in \mathbb{N}$  and  $n \geq 3$

by the Principle of Mathematical Induction.

5. (10 points) Prove, using the **precise definition for the limit of a sequence**. Do NOT use any of the algebraic limit theorems or any previous homework problems., that

$$\lim \sqrt{1 + \frac{1}{n}} = 1.$$

Please write your complete proof on the bottom half of the page labeled “**Proof.**”, only work that appears there will be graded. Your proof should be clearly laid out and your logic should be easy to follow. You may use the top half of the page labeled “*Discussion.*” for any scratch work. I may look at the scratch work if your proof is incorrect to help maximize the partial credit given.

*Discussion.*

$$\begin{aligned} \left| \sqrt{1 + \frac{1}{n}} - 1 \right| &= \left| \left( \sqrt{1 + \frac{1}{n}} - 1 \right) \left( \frac{\sqrt{1 + \frac{1}{n}} + 1}{\sqrt{1 + \frac{1}{n}} + 1} \right) \right| = \\ &= \left| \frac{1 + \frac{1}{n} - 1}{\sqrt{1 + \frac{1}{n}} + 1} \right| = \left| \frac{\frac{1}{n}}{n\sqrt{1 + \frac{1}{n}} + n} \right| = \frac{1}{n\sqrt{1 + \frac{1}{n}} + n} \\ &< \frac{1}{n} < \varepsilon \Rightarrow n > \frac{1}{\varepsilon} \end{aligned}$$

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**Proof.**

Let  $\varepsilon > 0$ . Set  $N = \frac{1}{\varepsilon}$ . Then, for  $n > N$  we have

$$\begin{aligned} \left| \sqrt{1 + \frac{1}{n}} - 1 \right| &= \left| \left( \sqrt{1 + \frac{1}{n}} - 1 \right) \left( \frac{\sqrt{1 + \frac{1}{n}} + 1}{\sqrt{1 + \frac{1}{n}} + 1} \right) \right| = \\ &= \left| \frac{\frac{1}{n}}{\sqrt{1 + \frac{1}{n}} + 1} \right| = \frac{1}{n\sqrt{1 + \frac{1}{n}} + n} < \frac{1}{n} < \frac{1}{N} = \frac{1}{1/\varepsilon} = \varepsilon. \end{aligned}$$

Therefore  $\lim \sqrt{1 + \frac{1}{n}} = 1$

6. (10 points) Prove the following statement, using the **precise definition for the limit of a sequence**. **Do NOT use any of the algebraic limit theorems or any previous homework problems.** If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $\lim s_n = c$  and  $\lim t_n = c$ , then  $\lim |s_n - t_n| = 0$ .

let  $\epsilon > 0$ . Since  $\lim s_n = c$ , there exists  $N_1$  such that for all  $n > N_1$ ,  $|s_n - c| < \frac{\epsilon}{2}$ . Since  $\lim t_n = c$ , there exists  $N_2$  such that for all  $n > N_2$ ,  $|t_n - c| < \frac{\epsilon}{2}$ .

Set  $N = \max\{N_1, N_2\}$ . Then for all  $n > N$

$$\begin{aligned} | |s_n - t_n| - 0 | &= |s_n - t_n| = |s_n - c + c - t_n| \leq |s_n - c| + |t_n - c| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

therefore  $\lim |s_n - t_n| = 0$ .